Exp. No. 4.4

Design and Simulation of 1-bit Comparator using Verilog code in Xilinx

Aim

To design and simulate a 1-bit comparator using Verilog HDL and verify its functionality using the Xilinx Vivado simulator.

Verilog Code

```
module comparator_1bit(
    input A,
    input B,
    output A_greater_B,
    output A_equal_B,
    output A_less_B
);
    assign A_greater_B = A & ~B;
    assign A_equal_B   = ~(A ^ B);
    assign A_less_B    = ~A & B;
endmodule
```

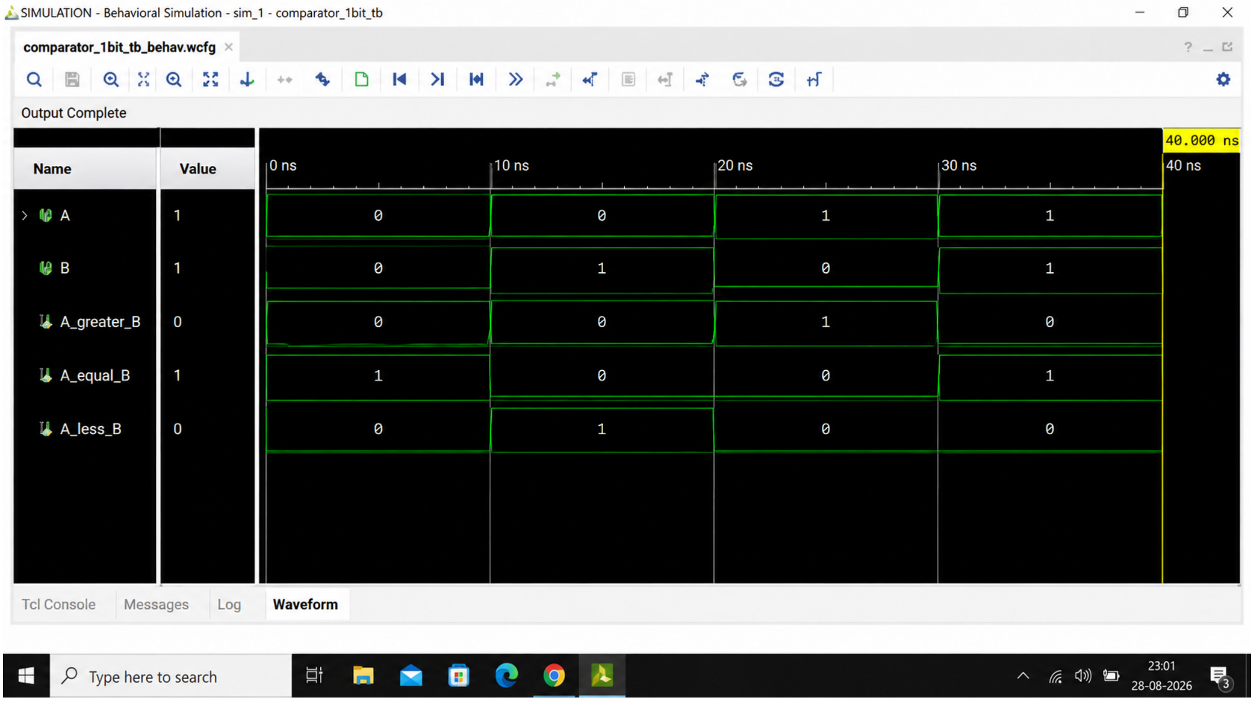
Testbench

```
`timescale 1ns / 1ps
module comparator_1bit_tb;
    reg A;
    reg B;
    wire A_greater_B;
    wire A_equal_B;
    wire A_less_B;

    comparator_1bit uut (
        .A(A),
        .B(B),
        .A_greater_B(A_greater_B),
        .A_equal_B(A_equal_B),
        .A_less_B(A_less_B)
    );

    initial begin
        A = 0; B = 0; #10;
        A = 0; B = 1; #10;
        A = 1; B = 0; #10;
        A = 1; B = 1; #10;
        $finish;
    end
endmodule
```

Output



Result: The 1-bit comparator was designed and simulated successfully in Xilinx Vivado. For A=0,B=0: A_equal_B=1; A=0,B=1: A_less_B=1; A=1,B=0: A_greater_B=1; A=1,B=1: A_equal_B=1, matching the expected truth table.